

LENYOLOSA EMMANUEL LENYOLOSA

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[LinkedIn](#) | [GitHub Portfolio](#)

PROFESSIONAL PROFILE

Recent Information Systems graduate (National University of Lesotho, B.Sc., 2026) with broad, hands-on experience across software development, database management, data analytics, and applications support. Comfortable working across the full stack — from database schema design and backend APIs to frontend interfaces and applied machine learning — and has independently built, deployed, and debugged multi-service systems end to end. Adaptable and fast-learning, with a demonstrated ability to move between technical domains (web development, data analysis, networking fundamentals) and to translate technical work into clear documentation and communication for non-technical audiences.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, PHP, SQL, C++

Web & Software Development: HTML5, CSS3, React, Django, Flask, Responsive Web Design, Object-Oriented Programming, REST API design & integration

Database & Data Management: MySQL, SQLite, database/schema design, SQL query development, data cleaning & transformation, database administration

Data Analytics & Machine Learning: Power BI, Pandas, NumPy, Matplotlib, Scikit-learn (classification & regression), exploratory data analysis (EDA), ETL pipeline development

Networking & Systems: TCP/IP fundamentals, computer communications and networks, cryptography and network security, computer organisation and architecture, operating systems fundamentals (coursework-based)

Tools & Platforms: Git & GitHub, Microsoft Excel, Microsoft Office Suite, Visual Studio Code, XAMPP/Apache, virtualization basics

Professional Skills: Requirements gathering & documentation, systems analysis and design, technical writing, troubleshooting & root-cause diagnosis, stakeholder communication, project planning

EDUCATION

Bachelor of Science in Information Systems

National University of Lesotho — Completed 2026 (Second Class, Second Division)

Relevant Coursework: Business Intelligence, Database Management Systems, Information Systems Development, Data Analytics, Systems Analysis & Design, Software Engineering, Project Management, Computer Networks, Computer Communications and Networks I, Cryptography & Network Security, Human-Computer Interaction, Management Information Systems

LGCSE — Leribe English Medium High School (2021)

JC — Leribe English Medium High School (2019)

PROFESSIONAL CERTIFICATIONS

- Data Analytics Essentials — Cisco Networking Academy, completed December 2025
- Introduction to Data Science — Cisco Networking Academy, completed August 2025

PROJECT PORTFOLIO

SmartSpend — Automated SMS-Based Personal Finance Tracker (Final Year Capstone) | *React, Flask, PHP, MySQL, Scikit-learn*

<https://github.com/Dilen-dev/smartspend>

- Co-designed and built a multi-service application (React frontend, Flask ML API, PHP REST backend, MySQL database) that ingests bank SMS notifications via webhook and automatically classifies and logs transactions.
- Implemented a three-layer classification pipeline (rule-based lookup, learned-store memory, and an SGDClassifier/TF-IDF model achieving 97.9% accuracy) to auto-categorise spending across 9 categories.
- Built a separate spend-forecasting model (SGDRegressor, R^2 0.809) with progressive cold-start predictions for new users and monthly incremental retraining via `partial_fit()`.
- Designed a relational MySQL schema (users, budgets, expenses, savings goals, category budgets) with foreign-key integrity, and implemented JWT-based authentication with protected routing.
- Authored formal requirements documentation, functional and non-functional specifications, and use-case analysis as part of the project proposal and final report.
- Diagnosed and resolved integration issues across the service chain (webhook delivery, database connectivity, API port conflicts) and configured a production deployment path (Apache virtual host, environment-based configuration).

Customer Churn Analysis | *SQL, Python, Power BI, Pandas, Scikit-learn*

<https://github.com/Dilen-dev/End-to-end-projects/tree/main/Data-Analysis/Churn-Analysis>

- Extracted, cleaned, and transformed customer data using SQL and Python, then ran exploratory analysis to surface behavioural patterns.
- Built a machine learning model to predict customer churn and classify at-risk customers.
- Designed interactive Power BI dashboards translating analysis into business-ready KPIs and retention recommendations.

HR Employment Analysis | *SQL, Power BI*

<https://github.com/Dilen-dev/HR-employment-analysis>

- Cleaned and prepared workforce data in SQL, standardising date fields and deriving new attributes (e.g. age from birthdate).
- Defined and answered 10 structured business KPI questions (turnover, tenure, headcount trends, departmental breakdowns) and built a Power BI dashboard to present findings.
- Produced a written insights report translating dashboard findings into HR strategy recommendations.

Premier League Data Pipeline (2022–2025) | *Python, REST API, Database*

<https://github.com/Dilen-dev/premier-league-2022-2025>

- Built an ETL pipeline that extracts data from a public REST API, transforms it, and loads it into a database for structured querying and analysis.

Online Banking Application | *Python, SQLite, Object-Oriented Programming*

<https://github.com/Dilen-dev/End-to-endprojects/tree/main/python/Online%20Banking%20Application>

- Developed a simulated banking application covering user registration, authentication, balance inquiries, deposits, withdrawals, and fund transfers.
- Applied object-oriented design to build modular, maintainable transaction-management features.

Payroll Management System | *C++*

<https://github.com/Dilen-dev/Payroll-management-system>

- Built a console-based application for employee record management, salary computation, and payslip generation, with login authentication and file-based data persistence.

Pharmacy Management System | *C++*

<https://github.com/Dilen-dev/pharmacy-management-system>

- Developed a console-based order and inventory management system using a singly linked list, supporting record edits, receipts, and daily sales summaries.

CORE COMPETENCIES

Software Development • Data Analytics • Database Management • SQL • Python • Applied Machine Learning • Problem Solving • Critical Thinking • Team Collaboration • Communication • Adaptability • Time Management • Attention to Detail • Continuous Learning